

Exam Date & Time: 22-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Makeup Examination - June -2024
Computer Vision [ECE-4051]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) For the 3x3 16-bit image $I(x,y)$ given below, compute the 3x3 output images of the Gaussian kernel $G(x,y)$ and the Laplacian kernel $L(x,y)$. Compare the output images of both the kernels and give applications where these kernels can be used.

A)

$$I(x,y) = \begin{bmatrix} 320 & 320 & 0 \\ 320 & 320 & 0 \\ 320 & 320 & 0 \end{bmatrix}$$

$$G(x,y) = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \quad (5)$$

$$L(x,y) = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

- B) Analyze the effect of sub-sampling an image with and without Gaussian LPF prefiltering. Justify which method is better and explain your reasoning. (3)
- C) Describe RGB color space with the help of a neat diagram. (2)
- 2) Analyze the strengths and limitations of Harris corner detector in detecting corners in an image. Explain whether Harris corner detector is translation, rotation, scale-invariant, and justify your answer. (5)
- A)
- B) Construct GLCM $C_{(0,1)}$ and $C_{(1,1)}$ for the given image. Discuss what information is obtained from these two matrices.

$$Img = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \quad (3)$$

- C) Explain the pooled texture representation using K-means algorithm. (2)
- 3) Design a segmentation algorithm to segment greenery (grass, trees, etc) in the following image. Clearly describe each step of the proposed algorithm. Discuss the possible limitations of the algorithm and the possible areas/regions which might not be accurately (5)

segmented. Justify your answer.

A)



B) Describe the correlation based approach to determine the pixel wise image correspondence (binocular fusion) from a pair of rectified images. (3)

C) Define Fundamental Matrix and explain its usefulness in binocular reconstruction (2)

4) Define shot in a video. Describe four different methods for shot boundary detection using interframe distance. (5)

A)

B) Differentiate between mean shift segmentation and k-means segmentation algorithms (3)

C) Explain in detail M estimator. Compare RANSAC with M estimators. (2)

5) Explain how decision boundary is determined by Support Vector Machine for classifying linearly separable data, with the help of equations. Compare linear kernel with non-linear kernel used in SVM. (5)

A)

B) A classifier is trained for quality check on an image dataset containing defective items (Class 1) and non-defective items (Class 0). While testing on a test set containing 1000 images of defective items and 1000 images of non-defective items, the classifier correctly classified 750 images of defective items and 800 images of non-defective items. Calculate precision, recall, and the confusion matrix of the test results. (3)

C) Explain k-fold cross-validation used for classification. (2)

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